

Dummy's Guide to Making a Loss Function

Notations:

$P(A)$: probability density function of A .

$P(A|B)$: probability density function of A , given B .

$A \sim N(0,1)$: A follows the standard normal distribution.

$f(x; \theta)$: function f of x with parameters θ .

Deriving MSE:

Let y, x represent some observed data.

Assume: $y = f(x; \theta) + \epsilon$ where $\epsilon \sim N(0,1)$

In plain english, we assume our response variable y is given by a function f of our predictor variable x , parametrized by θ with some slight noise.

Machine learning boils down to this question:

"What value(s) of θ are the most likely given the observed data x, y "

$$\operatorname{argmax}_{\theta} P(\theta | x, y) = ?$$

The process:

$$P(\theta|x,y) = \frac{P(\theta,x,y)}{P(x,y)}$$

definition of conditional probability

$$= \frac{P(y|x,\theta)P(x,\theta)}{P(y|x)P(x)}$$

$$= \frac{P(y|x,\theta)P(x|\theta)P(\theta)}{P(y|x)P(x)}$$

Assume θ is independent of x . If our model parameters don't change for each input. Therefore $P(x|\theta) = P(x)$

$$= \frac{P(y|x,\theta)\cancel{P(x|\theta)}P(\theta)}{P(y|x)\cancel{P(x)}}$$

$$= \frac{P(y|x,\theta)P(\theta)}{P(y|x)}$$

Since we're only interested in $\arg\max_{\theta}$ we can ignore terms that do not contain θ , namely $P(y|x)$.

$$\underbrace{P(\theta|x,y)}_{\text{'posterior probability'}} \propto \underbrace{P(y|x,\theta)}_{\text{'likelihood function'}} \underbrace{P(\theta)}_{\text{'prior probability'}}$$

The 'prior' represents our prior belief about θ before observing any data x,y . We update our prior using the 'likelihood' function, which describes the likelihood of observing data given θ . The 'posterior' is our updated belief about θ given the observed data.

To make our life easier later on, we'll apply $\log$ over both sides of the expression.

$$\begin{aligned}\log P(\theta | x, y) &\propto \log [P(y | x, \theta) P(\theta)] \\ &\propto \log P(y | x, \theta) + \log P(\theta)\end{aligned}$$

This is fine to do since $\arg\max_x f(x) = \arg\max_x \log f(x)$

Solving for log likelihood term

Recall that:

$$\begin{aligned}y &= f(x; \theta) + \epsilon \\ \epsilon &\sim N(0, 1)\end{aligned}$$

Since we are given x and θ , the only unknown random quantity is ϵ . In other words.

$$y \sim N(f(x; \theta), 1)$$

Let's notate $f(x; \theta)$ as $\hat{y}$.

$$\begin{aligned}P(y | x, \theta) &= \frac{1}{\sqrt{2\pi}} e^{-\frac{(y - \hat{y})^2}{2}} \\ \log P(y | x, \theta) &= \log\left(\frac{1}{\sqrt{2\pi}}\right) - \frac{1}{2}(y - \hat{y})^2 \\ &= -\frac{1}{2}(y - \hat{y})^2 \quad \text{ignore constant term.}\end{aligned}$$

This should look familiar to you — it's squared error!

Solving for log prior term

$P(\theta)$ represents our 'prior' belief about the distribution of θ . For instance, we may believe all values of θ are equally likely. AKA, we believe θ is uniformly distributed.

Assume $\theta \sim \text{Uniform}(a, b)$

$$P(\theta) = \frac{1}{b-a}$$

$$\log P(\theta) = -\log(b-a)$$

= constant that doesn't depend on θ

On the other hand, let's say we believe θ is normally distributed around 0, with some variance σ^2 .

Assume $\theta \sim N(0, \sigma^2)$

$$P(\theta) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(\theta-\mu)^2}{2\sigma^2}}$$

$$\begin{aligned}\log p(\theta) &= \log\left(\frac{1}{\sqrt{2\pi\sigma^2}}\right) - \left(\frac{\theta-0}{2\sigma^2}\right)^2 \\ &= -\left(\frac{1}{2\sigma^2}\right)\theta^2\end{aligned}$$

$$\text{Let } \lambda = \frac{1}{2\sigma^2}$$

$$\log p(\theta) = -\lambda\theta^2$$

This might look familiar to you — it's L2 regularization!

Conclusion:

Putting together the log-likelihood term and the log-prior term, we get:

$$\operatorname{argmax}_{\theta} \left[-\frac{1}{2} (Y - \hat{Y})^2 - \lambda \theta^2 \right]$$

$$\operatorname{argmin}_{\theta} \left[\frac{1}{2} (Y - \hat{Y})^2 + \lambda \theta^2 \right]$$

For multiple samples of x, y pairs we can generalize this like so.

$$\operatorname{argmin}_{\theta} \left[\sum_{i=1}^n \frac{1}{2} (Y_i - \hat{Y}_i)^2 + n \lambda \theta^2 \right]$$

To improve training stability and interpretability we normalize the expression by the batch size n .

$$\operatorname{argmin}_{\theta} \left[\frac{1}{2n} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2 + \lambda \theta^2 \right]$$

↑
Mean squared error
(from log-likelihood)

↑
L2 regularization
(from log-prior)